

ECE 501 - DIGITAL IMAGE PROCESSING
Explainable Multi-Label Content-Based Image Retrieval
for Aerial Images using Classical Computer Vision
Weekly Progress Report

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1. Introduction

During Weeks 2 and 3, the team focused on understanding the proposed Content-Based Image Retrieval (CBIR) system, studying the AID multi-label aerial image dataset, and planning the first implementation.

The main objective is to retrieve the most visually and semantically similar aerial images for a given query image using classical computer vision features. The complete system will use color, texture, shape, and spatial features, followed by multi-label semantic similarity.

The initial implementation focuses on **color-based retrieval** as a baseline.

2. Background Study

CBIR retrieves images based on their visual content rather than only using file names or text descriptions. An image is converted into numerical features, and these features are compared with the features of images in the database.

The dataset is multi-label, meaning that one aerial image can contain multiple scene categories or objects. Therefore, the final system will consider both visual similarity and semantic similarity.

For the first stage, color is used as the visual feature. A color histogram represents the distribution of colors in an image. RGB and HSV color spaces were studied, with HSV selected for the initial baseline because it separates color information from brightness more clearly.

The basic color pipeline is:

Image $\rightarrow$ HSV $\rightarrow$ Color Histogram $\rightarrow$ Normalized Feature Vector

3. Dataset Organization

The available dataset is organized into two folders:

- **images trained**: used as the gallery/database.
- **images test**: used as query images.

For each gallery image, a color feature will be extracted and stored. When a test image is selected as a query, its color feature will be compared with all gallery features and the most similar images will be retrieved.

4. Proposed System Flow

The complete planned system is shown in Figure 1.

The system will first extract handcrafted visual features from the gallery images. The same features will be extracted from a query image. Visual similarity will then be combined with multi-label semantic similarity to produce a final ranking.

The retrieved Top-K images will be accompanied by an explanation showing important feature contributions and, where possible, matching image regions. The system will also identify low-confidence or unreliable retrievals.

5. Color-Based Baseline

The first implementation will use color only.

The planned procedure is:

1. Load images from `images trained`.
2. Convert each image from BGR/RGB representation to HSV.
3. Calculate an H-S color histogram.
4. Normalize the histogram.
5. Store the histogram as the image feature vector.
6. Select an image from `images test` as the query.
7. Extract its color feature.
8. Compare the query feature with all gallery features.
9. Rank the images according to similarity.
10. Display the Top-K retrieved images.

This color-only system will serve as the baseline before adding texture, shape, spatial, and semantic information.

6. Work Completed

During Weeks 2 and 3, the following work was completed:

- Studied the concept and workflow of CBIR.
- Understood the multi-label nature of aerial images.
- Studied handcrafted visual features required for the project.
- Studied RGB and HSV color representations.
- Studied color histograms and histogram normalization.
- Examined and organized the available dataset.
- Defined `images trained` as the gallery and `images test` as the query set.
- Designed the complete CBIR system pipeline.
- Planned the first color-only retrieval baseline.

7. Plan for Next Week

- Implement HSV color histogram extraction.
- Extract color features from the gallery images.
- Implement similarity calculation.
- Retrieve and visualize Top-K results for test queries.
- Compare RGB and HSV based retrieval.
- Analyze the strengths and limitations of color-only retrieval.
- Begin implementation of texture features.

8. Conclusion

Weeks 2 and 3 established the foundation of the proposed aerial-image CBIR system. The team studied CBIR, multi-label similarity, and handcrafted visual features, and organized the dataset into gallery and query images.

A complete system pipeline was designed, and color was selected as the first feature for implementation. The next stage will produce the first working color-based retrieval results, which will serve as the baseline for the remaining features.

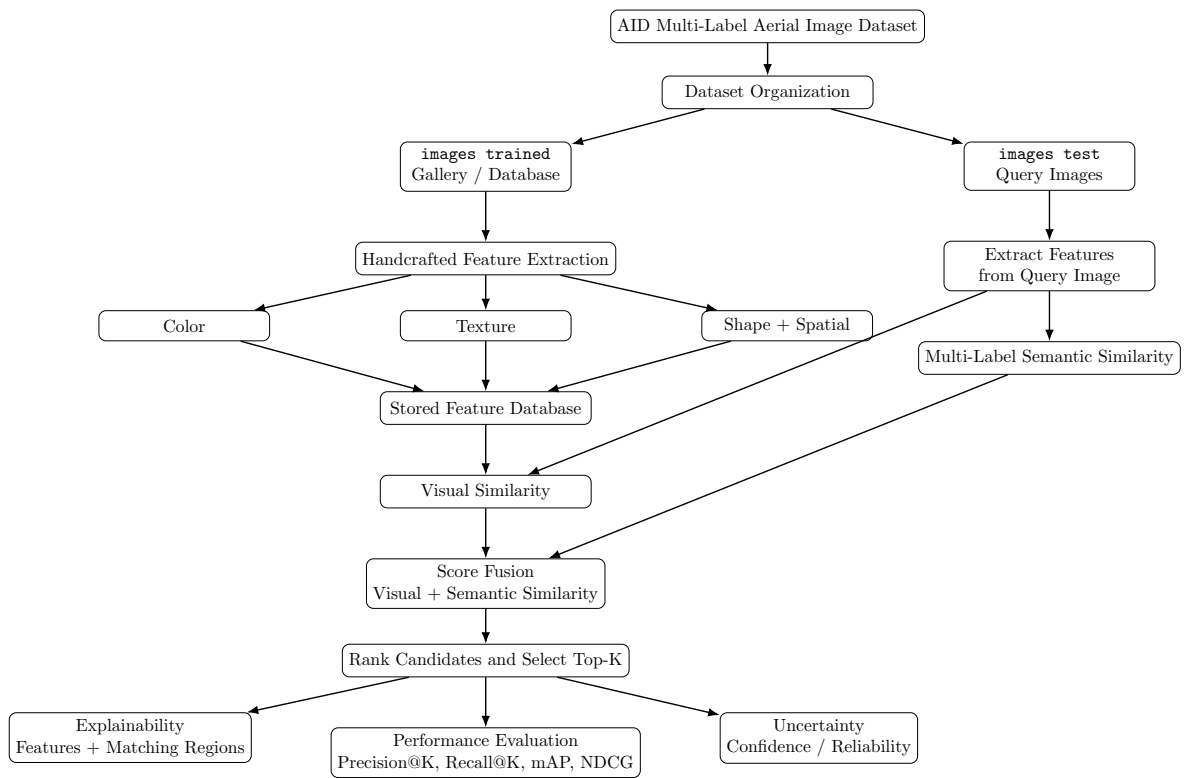


Figure 1: Proposed pipeline for explainable multi-label content-based image retrieval.